

CSS: Built-in Channel State Scrambling for Secure Wi-Fi based Sensing

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Abstract—This paper proposes CSS, a built-in channel state scrambling scheme to provide always-on protection for channel state information. CSS uses randomly generated scrambling vectors to emulate human activities, preventing eavesdroppers from recovering human physical activity from the channel state. By carefully designing the scrambling scheme based on physical channel models, we ensure that legacy receivers can successfully decode the frames and all frames transmitted over the air are scrambled. Furthermore, legitimate Wi-Fi sensors can still recover the true activity with a pre-shared secret key. Implementation on a real communication system shows that CSS can retain the same frame-error rate for commercial wireless receivers while misleading eavesdroppers with a success rate of over 95%.

I. INTRODUCTION

Wireless sensing systems utilize radio signals reflected by human body to achieve device-free sensing of human activities [1], [2]. One significant advantage of these sensing systems is their ability to integrate sensing and communication capabilities with widely used wireless protocols, such as Wi-Fi [3], [4], LTE [5], [6], RFID [7], and Lora [8], to enable a wide range of applications, such as fall detection [9], gesture recognition [10], [11], tracking [4], [12]. While wireless sensing has promising future applications, it also leads to potential privacy leakage, since private activity information is constantly broadcasted to the neighborhood in the form of wireless channel state variations. Eavesdroppers can capture these channel variations using the preamble of the frame without even decoding the content to derive privacy-sensitive information, such as keystrokes [13], [14], speech [15], vital signs [16], [17], and user identity [18], from a distance of tens of meters or even through the wall [2], [5]. Therefore, how to protect users from attackers that exploit wireless channel state as a side-channel for snooping privacy information becomes an important research issue.

Existing systems introduce artificial channel fluctuations to protect users from activity information leakage. One approach is to obfuscate the channel using third-party wireless devices, such as physical layer relays [19], [20], Intelligent Reflection Surfaces (IRS) [21], or motorized hardware reflectors [22]. However, deploying extra hardware in the target area is necessary for these obfuscating systems. Thus, it is difficult to cover a large region or protect a user carrying the mobile transmitter

while moving around. The second approach is to obfuscate the wireless signal using the transmitter itself by randomizing frame rate/transmission power [23] or adding extra phase or amplitude noises to the transmitted signal [24]–[26]. These obfuscating schemes are tailored to specific applications, such as gesture recognition [24], [27], without considering general protection for daily-life human activities.

In this paper, we propose CSS, a built-in Channel State Scrambling scheme to provide always-on protection for channel state information (CSI). Our end-to-end scrambling scheme can be implemented as a lightweight modification to the Network Interface Card (NIC) driver and physical layer (PHY) of Wi-Fi transmitters. The driver generates random scrambling vectors to emulate channel state variations caused by human activities. The PHY then uses the scrambling vectors to alter the transmitted symbols, in a way similar to traditional beamforming. Therefore, eavesdroppers will perceive irrelevant activities even if there are no environmental movements.

To enable built-in channel state scrambling, we need to address the following three technology challenges in CSS. First, CSS should not degrade the communication performance of legacy Wi-Fi devices. To address this challenge, we carefully control the scramble parameters used in CSS according to the wireless channel model so that the amplitude and phase of the scrambled signal are consistent with the physical channel variations. Second, to ensure effective obfuscation, CSS should be able to generate a large number of random human activity patterns. To address this challenge, we design a Linear Predictive Code (LPC) based scheme to generate a random set of scrambling vectors with similar signal patterns to general human activities that provides better camouflage performance than random noises. Third, CSS should allow legitimate sensing devices to recover the real activity signals. To achieve this, we design a pseudo-random scrambling vector scheme so that the legitimate receiver can use a shared secret key to descramble the received signal.

We implemented CSS using the OpenWiFi framework [28] on the Xilinx ZCU-102 FPGA platform. Our experiments on commercial Wi-Fi receivers show that the scrambling does not affect the communication performance between CSS enabled devices and legacy receivers. Our evaluations with real human activities show that our scrambling scheme can mislead the attacker with a success rate of over 95%, while legitimate

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sensors are able to recover activity signals.

In summary, we have made the following contributions.

- We are the first to use the LPC algorithm for generating random non-stationary signals that resemble human activity signals.
- We propose and implement a general Wi-Fi scrambling framework that provides always-on protection for channel state information. CSS manipulates the artificial channel state in frequency domain with a time granularity of 0.1 milliseconds in the user space of the operating system, which allows our framework to be easily extended to support other scramble algorithms, *e.g.*, protection of tracking or gesture recognition.
- We develop a descramble algorithm for legitimate sensing devices that have a secret key that used by the transmitter to recover the real channel state.

II. RELATED WORK

Security and privacy issues in Wi-Fi sensing: Recent research shows that attackers can capture the wireless channel state without decoding the encrypted frame content [29]. Leaked channel states may reveal sensitive privacy like keystrokes, speech, vital signs, and user identity due to its high correlation with surrounding movements. WiKey captures subtle keystroke movements and recover passwords using commercial Wi-Fi devices [13]. WiHear recovers human speech by lip-reading [15]. In these cases, the wireless signal leaks highly-sensitive human movement information to a distance of tens of meters or even through the wall [30], [31]. Attackers can also extract breath and heartbeat [16], [17] signals or identify the user by measuring the gait features using wireless signals [18], [32], [33]. Therefore, it is important to provide a ubiquitous protection scheme for ordinary Wi-Fi users.

Passive protection schemes: When the attacker actively sends wireless signals to collect channel information, it is possible to interfere it by sending signals in adjacent channels [34]. Zhang *et al.* propose redirecting the user to a noisy environment when the user wants to perform a private action [35]. Ghostbuster proposes a detection scheme for analyzing the patterns of Radio Frequency (RF) leakage of different types of devices [36]. In addition, EarFisher detects the Electromagnetic Radiation (EMR) of the eavesdropper's memory access [37]. However, both detection schemes require specially designed devices and have limited detection ranges.

Active obfuscating schemes: PhyCloak proposes to use a duplex relay node to introduce obfuscating multipath signals [19]. Aegis uses physically rotating objects to provide CSI obfuscation within the range of a house [22]. IRShield uses intelligent reflection surfaces to produce the obfuscating signal [21]. RF-protect uses switches and phase shifters to emulate human trajectories under FMCW radars [20]. However, these third-party obfuscating schemes require extra hardware.

The most related approaches to our work are systems that use transmitter-based obfuscating. Xu proposes to alternate the preamble phase to hide the CSI phase information [38]. The IEEE standard 802.11az also introduces randomized phase shifts to protect ranging frames [39]. Jiao *et al.* introduce extra

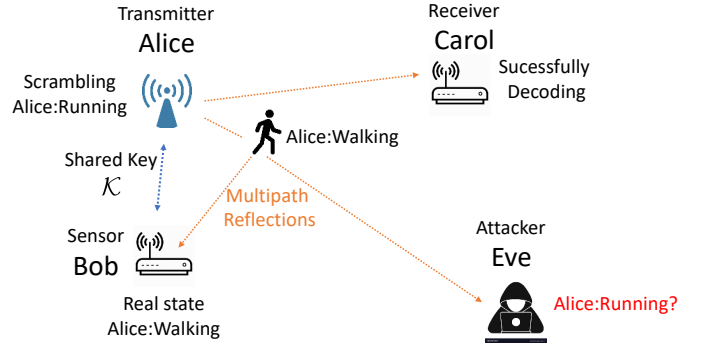


Fig. 1. Attack scenario

channel impulse response (CIR) taps in the physical layer using the OpenWi-Fi platform [25]. RF-Veil introduces random phase shifts along the frequency domain to protect device RF fingerprints [40]. Cominelli *et al.* introduce random filters to preserve user's location privacy [41] [42]. However, these obfuscating schemes do not consider the specific time domain variations of human activities, so they serve as random-noise-based obfuscations that are less efficient than emulated human activities. And their implementation in real system is inadequate for emulating fine-grained signal variations induced by moving objects. WiAdv obfuscates the signal with Doppler patterns to mislead human gesture recognition systems [27]. MoRTr emulates human gestures by replaying randomized gesture signals captured in the same environment [24]. Furthermore, legitimate Wi-Fi sensors may not be able to recover the real CSI from the obfuscated signal in these systems.

III. SYSTEM OVERVIEW

In this section, we first describe the threat model and the assumptions we made. We then introduce the system framework of CSS.

Threat Model: As shown in Fig. 1, we assume that the Wi-Fi transmitter, Alice, needs to transmit a certain amount of information to a set of receivers. These receivers may have different roles, such as being Wi-Fi sensors (*e.g.*, Bob), who are legitimate receivers that can recover the real channel state, or being legacy receivers (*e.g.*, Carol), who only need to decode the content of the frames. To conform with standard wireless protocols, Alice must include predefined preamble and pilot signals in the frame to ensure successful frame decoding at the receivers. However, Alice does not want the physical movements around her to be discovered by unauthorized parties. Only legitimate Wi-Fi sensors that have the shared secret key $\mathcal{K}$ can recover the real channel state from the transmission.

We assume the attacker, Eve, deploys a wireless signal monitor with one antenna to capture the transmission from Alice by passively listening to the wireless channel. Eve has full knowledge of the wireless standard, including the format of preambles and pilot signals, allowing her to measure the channel state on multiple subcarriers of the Orthogonal Frequency-Division Multiplexing (OFDM) signal transmitted by Alice. The objective of Eve is to recover the activity infor-

mation of Alice's movements from the captured channel state. Additionally, Eve may be aware of the scheme used by Alice for scrambling the frame, including the activity templates used for scrambling. However, Eve has no knowledge of the secret key $\mathcal{K}$. We assume that Eve can store the captured signal for offline analysis. We further assume that Eve does not actively transmit wireless signals, because a nearby transmitter can be easily discovered or localized by Alice [36]. We do not assume that Bob and Carol are closer to Alice than Eve, nor the channel between Alice and Eve is correlated with other channels. Extension for scenarios with multiple transmitters and attackers with multiple antennas will be further discussed in Section IV-E.

System Design: The overall design of CSS for the transmitters and legitimate Wi-Fi sensors is shown in Fig. 2. The transmitter first uses the shared secret key $\mathcal{K}$ to generate the scrambling vector sequence $S(f, t)$, where f is the frequency of a subcarrier and t is the timestamp of the frame. The exchange of secret key $\mathcal{K}$ can be done after Wi-Fi Protected Access (WPA) authentication [43], the detail is omitted here. The per-frame scrambling vector contains s complex-valued elements, one for each subcarrier used by the Wi-Fi frame. The scrambling vector generation algorithm consists of two components. First, for the time dimension, we use a LPC-based random signal generator that emulates the Doppler spectrogram of human activities. Second, for the frequency dimension, we use the channel delay model to emulate the phase and amplitude changes among different subcarriers. The transmitter multiplies the scrambling vector with each symbol of the modulated frame content in the frequency domain. Subsequently, the transmitter performs an inverse Fourier transform to convert the signal into the time domain and transmits it through the wireless channel as a normal Wi-Fi transmitter.

The receiver first decodes the received frame using normal channel estimation and decoding schemes. As we will show in later discussions, legacy receivers can decode the frame content without being aware of the scrambling process. However, the channel estimation will be obfuscated by the scrambling vector introduced by the transmitter. To recover the real channel state and perform Wi-Fi sensing tasks, the receiver need to use the shared key $\mathcal{K}$ along with the timestamp corresponding to the frame to generate the same scrambling vector used by the transmitter. By dividing the fake channel estimation with the scrambling vector, the receiver can recover the real channel state that reflects the surrounding movements.

IV. CHANNEL STATE SCRAMBLING

In this section, we describe how CSS generates the scrambling vectors and the mechanism for secure sensing.

A. Channel Scrambling Model

Wi-Fi protocols, such as IEEE 802.11n/ac/ax, use OFDM modulation schemes that divide the channel into multiple subcarriers. On each subcarrier with a frequency of f , the transmitter transmits a signal $X(f, t)$ at time t , and the received signal is:

$$Y(f, t) = H(f, t) \times X(f, t) + N(f, t) \quad (1)$$

where $H(f, t)$ is the Channel Frequency Response (CFR), and $N(f, t)$ is the noise. Note that we only focus on a single pair of transmitter/receiver antennas and ignore the noise $N(f, t)$ in the later part of this paper for simplicity.

A common model [3] used for Wi-Fi sensing is to represent the CFR as:

$$H(f, t) = H_s(f) + \sum_p A_p(f, t) e^{-j \frac{2\pi d_p(t) f}{c}} \quad (2)$$

where $H_s(f)$ is a static component that does not change with time. The component $\sum_p A_p(f, t) e^{-j 2\pi d_p(t) f / c}$ represents the dynamic multipath introduced by human activities, where p is a signal propagation path, $A_p(f, t)$ is the complex-valued attenuation for path p , and $d_p(t)$ is the length of path p .

When the receiver knows the transmitted signal $X(f, t)$, it can estimate the CFR $H(f, t)$ by dividing the received signal $Y(f, t)$ with $X(f, t)$. By analyzing the changes in the amplitude or phase of $H(f, t)$, the receiver can recover the Doppler shift or path distance change that is directly related to surrounding human movements. In this way, Wi-Fi sensors can detect subtle movements, such as breathing or keystrokes, and recognize different types of movements with high accuracy.

To obfuscate the channel state, CSS scrambles the transmitted signal by a scrambling vector sequence $S(f, t)$:

$$X'(f, t) = S(f, t) \times X(f, t) \quad (3)$$

Therefore, the received signal $Y'(f, t)$ will be $Y'(f, t) = H(f, t) \times S(f, t) \times X(f, t)$. With the knowledge of $X(f, t)$ only, the receiver will get the CFR estimation as:

$$H'(f, t) = H(f, t) \times S(f, t) \quad (4)$$

By manipulating the scrambling vector $S(f, t)$, the transmitter can control the CFR $H'(f, t)$ perceived by the receiver. Consider the case where the CFR only has a constant static component of $H_s(f)$ for a static scenario with no movements. If the transmitter wants to emulate a target CFR of walking $H_w(f, t) = H_s(f) + \sum_{p_w} A_{p_w}(f, t) e^{-j \frac{2\pi d_{p_w}(t) f}{c}}$, which represents a user walking around, it should set:

$$S_w(f, t) = 1 + \sum_{p_w} A_{p_w}(f, t) e^{-j \frac{2\pi d_{p_w}(t) f}{c}} \quad (5)$$

where p_w are the paths affected by walking. In this case, the receiver will perceive a CFR of:

$$H_w(f, t) = H_s(f) + H_s(f) \sum_{p_w} A_{p_w}(f, t) e^{-j \frac{2\pi d_{p_w}(t) f}{c}} \quad (6)$$

which is the static CFR superimposed with a dynamic component similar to the walking activity. Therefore, the eavesdropper will detect an activity emulated by the transmitter even in a static environment. When the original CFR is already affected by the physical movements of surrounding users, the scrambling scheme will superimpose another activity on the existing activity. The eavesdropper will perceive a mixed signal and cannot tell what it is.

Ensuring legacy communication: Legacy Wi-Fi receivers usually use the Short Training Field (STF), Long Training

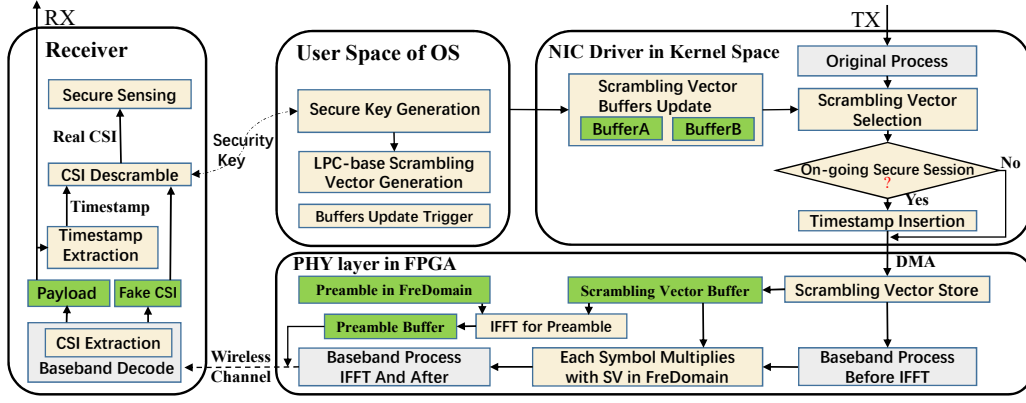


Fig. 2. System design and implementation

Field (LTF), and pilot subcarriers to estimate the channel state, *i.e.*, $H(f, t)$, and decode the frame based on $H(f, t)$. After scrambling, the receivers would perceive the scrambled channel state $H'(f, t)$ instead of the real $H(f, t)$.

To ensure correct decoding, we should consistently apply the scrambling vector as if the changes in the physical world truly generate these artificial channel variations. First, the scrambling vector should be applied in the same way to all symbols in a frame, including the STF, LTF, and data symbols. Thus, the receiver gets a consistent estimation of channel state as $H'(f, t)$ and then recover $X(f, t)$ by $X(f, t) = Y'(f, t)/H'(f, t)$. Second, the time domain variations should conform with the channel coherence time requirements of Wi-Fi protocols, which limit the maximum movement speed and Doppler frequency of transceivers. In CSS, we set the scrambling vector to be the same for each frame, so there is no artificial channel state variation within a single frame. For inter-frame time dependencies in the scrambling vector, see details in Section IV-B. Third, the frequency domain variations should conform with the coherence bandwidth requirements, which limit the maximum delay spread of signals. To address this problem, we design a scheme to smooth the scrambling vector along the frequency domain, see details in Section IV-C.

B. Time Dimension Scrambling

Human activities generate time dimension variations in the channel state in the form of Doppler shifts and amplitude/phase variations. Therefore, the key to emulating the channel state generated by human activities is to simulate the time-domain CFR waveforms. To simplify our discussion, we focus on a single subcarrier, the center subcarrier with a frequency of f_0 , in this section¹. The variation in the frequency domain will be further discussed in Section IV-C.

1) *Time Dimension Channel Models*: As discussed in Section IV-A, when we wish to emulate a walking activity, the scrambling vector $S(f_0, t)$ should be set according to Eq. (5). We observe from Eq. (5) that the value of the scrambling is a constant of 1, which represents the static path, plus

¹Note that the center subcarrier is not affected by the Sampling Frequency Offset (SFO). Normally, there is no signal transmitted in the center subcarrier, so it is just a “virtual” subcarrier.

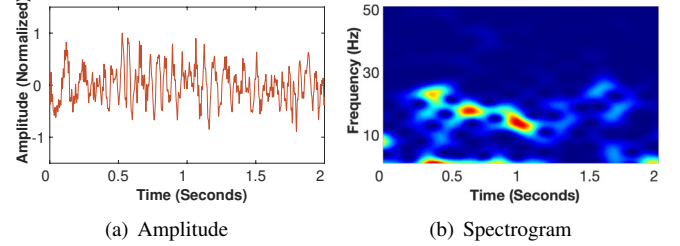


Fig. 3. CFR waveform and spectrogram for walking

the summation of complex-valued changes caused by activity movements. Research in Wi-Fi-based activity recognition has shown that the changes caused by activities are mainly due to a slow amplitude/phase change in $A_{pw}(f, t)$ and a fast Doppler shift in $e^{-j\frac{2\pi d_{pw}(t)f}{c}}$ due to changes in path length $d_{pw}(t)$ [3]. Therefore, the CFR waveform for a given activity combines multiple time-varying frequency components. Fig. 3 shows the typical amplitude waveform and spectrogram for a walking activity captured at the 5GHz band. We observe a rough frequency band around 25 Hz that corresponds to the Doppler shift associated with walking speed.

A straightforward way to emulate time variations for human activity is to record sample templates for each activity and replay these templates during the transmission of Wi-Fi frames. However, continuously replaying a fixed number of templates could be easily detected by the attacker. Moreover, to enable secure channel state information recovery discussed in Section IV-D, the templates should be public for legitimate CSI sensors to remove the scrambling. In this case, the attacker could launch a brute-force attack by guessing which template the transmitter is using and perform the same recovery process as legitimate sensors when the number of templates is small.

We use a secret key $\mathcal{K}$ to generate per-frame scrambling vectors to provide enough randomness in the scrambling vectors. Our system uses the LPC framework along with the random secret key to generate a non-stationary scrambling vector sequence.

The secret key introduces randomness in the following aspects of the scrambling vector:

- The number of components in the scrambling vector and the weight of each component, see the following discussion.
- The LPC coefficients and the LPC excitation used for

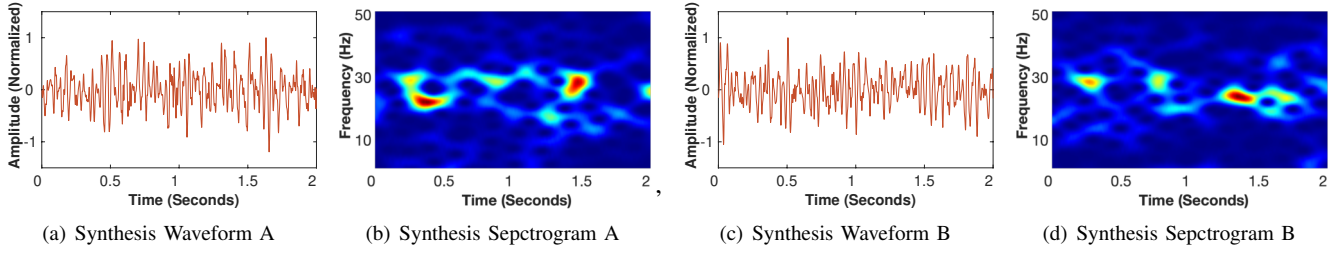


Fig. 4. LPC synthesised waveforms and spectrograms using the same walking coefficients.

each component, see Section IV-B2.

- The activity duration and state transitions for non-stationary activities, see Section IV-B3.
- The variation across the frequency dimension (subcarriers), see Section IV-C.

Therefore, for a key with a length of 1024 bits, there would be 2^{1024} different sets of scrambling vectors, so the attacker can hardly guess the scrambling vector without the key.

Our scrambling vector can be represented as:

$$S(f, t) = 1 + \sum_{i=1}^m \lambda_i D_i(f, t) \quad (7)$$

which is the summation of m time-varying components. The scrambling intensity, λ_i , is the weight for component i that controls the magnitude of variation generated by the component. The dynamic component $D_i(f, t)$ is the normalized artificial variation that resembles the variation generated by human activity, i.e., $D_i(f, t)$ is a complex vector sequence that has a similar spectrogram as $\sum_{p_w} A_{p_w}(f, t) e^{-j \frac{2\pi d_{p_w}(t)f}{c}}$. Therefore, the resulting $S(f, t)$ is a weighted combination of different components, and each represent a subject with different activities. The magnitude of each component is normalized so that the maximum of $|D_i(f, t)|$ is one. Therefore, by changing the value of λ_i , we can control the amplitude of variations in $S(f, t)$. For example, if we have only one component, we set λ_1 to 0.05. Then, the magnitude of $S(f, t)$ is bounded between $[0.95, 1.05]$.

2) *LPC-based Waveform Synthesis*: The dynamic components for human activities consist of multiple time-varying frequency components, as shown in Fig. 3(b) and Eq. (5). We propose to use the LPC framework to generate random non-stationary signals that resemble human activity signals.

The LPC scheme has been widely applied in speech signal processing for speech coding and speech synthesis [44]. We observe that human activity signals have similar characteristics as speech signals. They are both non-stationary and contain both periodic segments, such as voiced sounds in speech (e.g., vowels) or periodic activities (e.g., walking), and non-periodic segments, such as voiceless sounds (e.g., plosive sounds) in speech or short activities (e.g., keystroke). The characteristics of both signals are determined by their frequency components in short signal segments and the transitions between segments. Therefore, we choose to use LPC as our time-dimension-based waveform synthesis scheme.

The signal synthesis process for LPC is as follows. First, the input $e(n)$ is an excitation signal sequence, a periodic impulse train for periodic segments, or random noise for non-

periodic segments. Next, the excitation $e(n)$ passes through the LPC filter, which contains a short-term feedback predictor, to generate the output $d(n)$:

$$d(n) = \sum_{k=1}^L a_k d(n-k) + e(n) \quad (8)$$

where a_k are the coefficients for the short-term predictor and L is the order of the LPC filter.

The coefficients of the predictor, a_k , determine the spectral density of the output signals. Therefore, we need to extract the LPC coefficients a_k from the waveforms of real-world activities to synthesize an output similar to human activity signals. We first collect CSI waveforms while users perform the given activity, then pass them through a general CSI processing pipeline, including denoising, DC removal, and segmentation. Note that, we only need to collect the real CSI in one environment, since what is needed is its dynamic component. We use the standard autocorrelation method [45] to extract the real-valued LPC coefficients from the amplitude of the denoised CSI waveforms. Our experiments show that with a segment length of one second, a sampling rate of 800 and a predictor order of 50, i.e., $L = 50$, CSS can provide a good approximation for most activities signals. Therefore, we extract a set of 50 coefficients for each segment. Note that these LPC coefficients serve as compact representations of spectrograms so that users can define customized patterns and share the patterns with legitimate Wi-Fi sensors.

For a fixed set of LPC coefficients, we can use different random excitation signals to generate diverse outputs. Fig. 4 shows two synthesized output examples using the same walking LPC coefficients. We observe that both the waveform and spectrogram of the synthesized signals are similar to the real activity samples in Fig. 3(a). Therefore, it is hard for the eavesdroppers to determine whether these signals are real or artificial, as we will show in our evaluations in Section VI. Nevertheless, we can observe that these synthesized signals exhibit differences in detail. Even if these synthesized signals are derived from the same LPC coefficients, the outputs differ due to their different excitation. This enables us to generate a large number of random samples for the same activity so that the eavesdroppers cannot guess the exact scrambling vector selected by the transmitter. In this way, we can get a random scrambling sequence that is generated by the secret key $\mathcal{K}$ using a deterministic process.

The output of LPC, $d(n)$ is a real-valued discrete sequence with a fixed sampling rate of 800 samples per second. To convert it to the complex-valued $D(f, t)$, we first use the

Hilbert Transform to recover its imaginary component. Then, we interpolate the sequence to a higher sampling rate of $10KHz$ so that the transmitter can select a suitable scrambling vector using a transmission timestamp with a time granularity of 0.1 milliseconds. The transmission timestamp indicates the elapsed time from the beginning of a given scramble session, and the session identifier is shared between the transmitter and the legitimate Wi-Fi sensors.

LPC-based vs. Random-Noise-based: The synthetic signal generated by LPC is more effective than the traditional random-noise-based approach. Because random white noise distributes its energy evenly across the whole frequency range. Therefore, it fails to conceal human-induced Doppler shifts that are concentrated in a narrow frequency range. Fig. 5(a) shows the spectrogram of sit-down (standing after sitting). Fig. 5(b) and 5(c) show the spectrograms of sitting down obfuscated by white noise and LPC of walking in the same scrambling intensity. We can observe that the spectrum pattern of the sit-down activity is apparent in the white-noise-based scrambling, but indistinct in the LPC-based scrambling of walking activity.

3) *Non-stationary Activity Generation:* We use the secret key $\mathcal{K}$ to generate pseudo-random numbers to determine the activity state transitions in a Markov chain. Each activity state is represented by a set of LPC coefficients. Therefore, the output state sequence is still deterministic with the key $\mathcal{K}$.

C. Frequency Dimension Scrambling

In the previous section, we have discussed channel state scrambling in the time dimension. As Wi-Fi devices transmit in multiple OFDM subcarriers simultaneously, the eavesdropper may measure the channel state on different subcarriers. Thus, it is important to consider scrambling along the frequency dimension, since applying the same scrambling value to all subcarriers may not provide enough protection.

Consider the case that the eavesdropper measures the channel state on two subcarriers, *e.g.*, f_1 and f_2 . When we use the same scrambling value $S(t)$ for all subcarriers, the scrambled channel state measured by the eavesdropper will be:

$$H'(f_1, t) = H(f_1, t) \times S(t) \quad (9)$$

$$H'(f_2, t) = H(f_2, t) \times S(t) \quad (10)$$

If the eavesdropper divides the two-channel measurements and the scrambling introduced by $S(t)$ can be canceled:

$$\frac{H'(f_1, t)}{H'(f_2, t)} = \frac{H_s(f_1) + \sum_p A_p(f_1, t) e^{-j \frac{2\pi d_p(t) f_1}{c}}}{H_s(f_2) + \sum_p A_p(f_2, t) e^{-j \frac{2\pi d_p(t) f_2}{c}}} \quad (11)$$

As the frequency for $H(f_1, t)$ and $H(f_2, t)$ are different, their ratio will contain residual information about the dynamical changes in the channel state in a way similar to the CSI-ratio approach [16]. To see this, we can multiply both the denominator and the numerator by the conjugate of $H(f_2, t)$, which is $H^*(f_2, t) = H_s^*(f_2) + \sum_p A_p^*(f_2, t) e^{j \frac{2\pi d_p(t) f_2}{c}}$. We observe that when there is no movement in the environment, $H(f, t) = H_s(f)$, the ratio in Eq. (11) remains constant, $H_s(f_1)H_s^*(f_2)/|H_s(f_2)|^2$, along the time dimension. Therefore, the eavesdropper can determine that there is no movement

even if the transmitter has scrambled the signals. When there are movements, we observe that $\frac{H'(f_1, t)}{H'(f_2, t)}$ will contain terms proportional to $H^*(f_2, t) \sum_p A_p(f_1, t) e^{-j \frac{2\pi d_p(t) f_1}{c}}$. These terms are distorted versions of the physical movement. Fig. 5(c) shows the scrambled version of a real-physical periodic sitting down activity with a fake walking activity. The scrambling process successfully adds high-frequency components of walking activity into the spectrogram. However, the CSI-ratio approach removes most of the high-frequency scrambling and retains the sitting down components in the spectrogram, as shown in Fig. 5(d).

A complete random scrambling vector along the frequency dimension without considering physical channel models may lead to two problems. First, if the amplitude and phase are not correlated along the frequency dimension in a way that matches physical human movements, the attacker may be able to infer that the channel variations are artificial. Second and more importantly, if the frequency dimension variations are inconsistent with the channel coherence assumptions of Wi-Fi protocols, the receiver may not be able to decode the frame successfully. Most receivers rely on the training sequence, such as cross-correlation of LTF, to achieve frame synchronization. However, when the scrambling vector is completely random along the frequency dimension, the receiver may miss the cross-correlation peak between the known training sequence and the received signals. As a result, the receiver may incorrectly determine the starting point of a frame and be unable to decode the frame. We have verified this using our implementation discussed in Section V. When the phases of scrambling vectors are random along the frequency dimension, the commercial NICs cannot even detect the Service Set Identifier (SSID). That indicates the receiver is unable to decode scrambled frames.

We construct the frequency domain scrambling vector based on the synthesized dynamical component at the center subcarrier. The scrambling vector can be represented as:

$$S(f, t) = e^{-j2\pi\tau_0(t)f} + \sum_{i=1}^m \lambda_i D_i(f_0, t) e^{-j2\pi(\tau_0(t)f + \tau_i(t)(f-f_0))}. \quad (12)$$

Compared to the simplified representation in Eq. (7), we further randomize the phase of each component using two types of delays. The first type of delay, $\tau_0(t)$, is a time-varying delay applied to all components, including the static component. This delay introduces a linear phase shift of $2\pi\tau_0(t)f$ for different subcarriers, similar to the effect of the Sampling Frequency Offset (SFO) in CSI. By slowly changing the values of $\tau_0(t)$ as time evolves, we can emulate a transmitter slowly moving around or having a random SFO. The second type of delay, $\tau_i(t)$, is specific to each dynamic component. By changing the $\tau_i(t)$ of each component, we can emulate a moving subject with a constant or randomly varying change in distance.

We observe from Eq. (12) that by introducing the delay factors, the phase of each component in $S(f, t)$ is dependent on the frequency f since different subcarriers are transmitted at different frequencies. Therefore, dividing the measurements

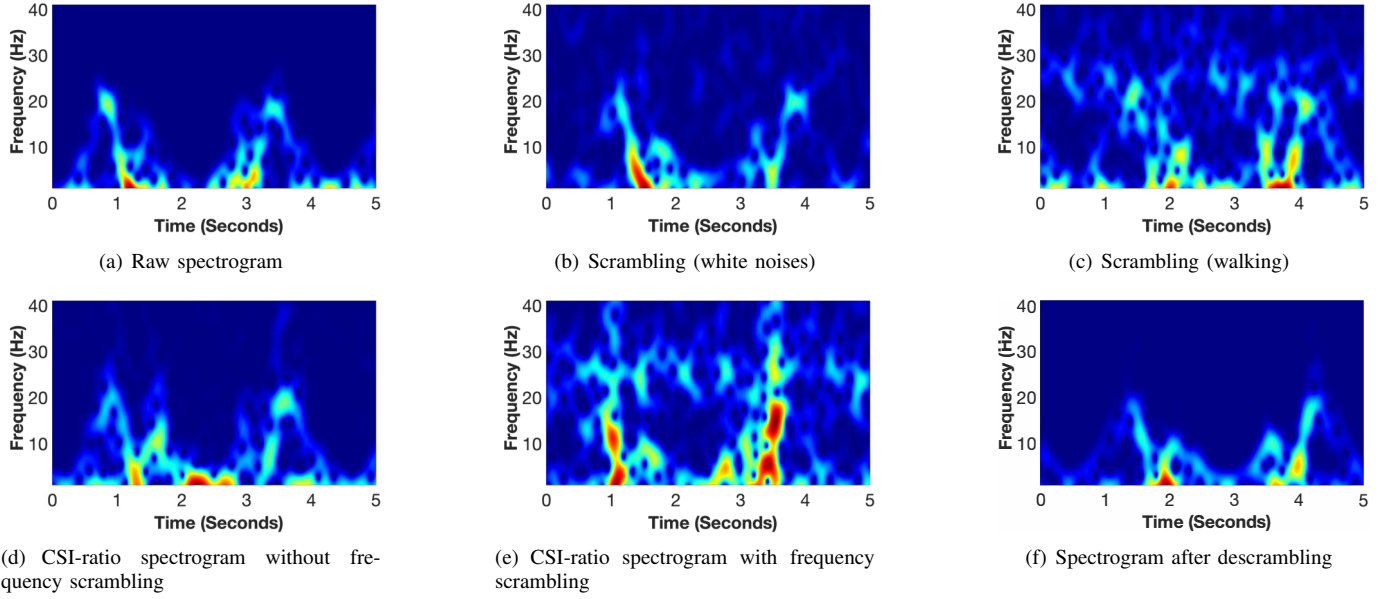


Fig. 5. Spectrogram of a sit-down activity scrambled by walking.

as in Eq. (11) cannot cancel $S(f_1, t)$ with $S(f_2, t)$ because they are combinations of dynamic components with different phases. Detailed derivations are omitted due to space constraints. To illustrate the performance of frequency dimension scrambling, we show the CSI-ratio spectrogram when frequency domain scrambling is applied in Fig. 5(e). In this case, we observe that the CSI-ratio is no longer able to remove the high-frequency components of the fake-walking activity.

We impose two additional constraints on these delays to ensure their values conform with physical layer models. First, the value of $\tau_i(t)$ should be within a range of $[0, \tau_{max}]$. Furthermore, we need to limit the maximum delay spread τ_{max} between the dynamic component and the static component according to the typical delay spreads in WLAN [46]. Otherwise, the excess delay spread may lead to errors in the demodulation process. We set the maximum delay spread to be $100ns$, which is half of the typical delay spread, to provide room for extra delay spreads introduced by the real physical channel. Second, the time variations of $\tau_i(t)$ should be within the channel coherence time limits for WLAN. As the movement of the reflection path causes the changes in $\tau_i(t)$, we can change $\tau_i(t)$ based on the activity information. For example, if we are emulating a walking process, we can set $\tau_i(t)$ as a linear function of time t according to the walking speed, *e.g.*, $0.8 - 1.2m/s$. The above frequency dimension scrambling also changes the overall signal strength so that attackers who use the received signal strength indicator (RSSI) as the feature would also perceive similar activities as the CSI.

The initial values and time variations of the random delays $\tau_0(t)$ and $\tau_i(t)$ can be generated by the key $\mathcal{K}$. We can use the timestamp to generate the full set of $S(f, t)$ that varies along the time and frequency dimensions using Eq. (12). The transmitter will use the transmission timestamp t to determine the scrambling vector $D(f, t)$ for a given Wi-Fi frame and apply the scrambling vector to all symbols of the frame.

D. Secure CSI Sensing

Besides the randomness aspects described in IV-B1 that determine the parameters of LPC-based signal generator, the key $\mathcal{K}$ also indicates the begin time of a secure sensing session. By virtue of the LPC-based signal generator, the scrambling vector is deterministic with the shared key $\mathcal{K}$ and the timestamp of the frame. And the Timing Synchronization Function (TSF) is capable of providing time synchronization with microsecond-level precision [47]. Moreover, the output rate of scrambling vector is constant, *i.e.*, $10KHz$, so legitimate sensors can generate the same set of scrambling vectors of $S(f, t)$. With the regenerated scrambling vector $S(f, t)$, the sensing receiver can recover the real channel state by $H(f, t) = H'(f, t)/S(f, t)$. Fig. 5(f) shows a recovered sitting down spectrogram of its scrambled counterpart, *i.e.*, Fig. 5(c).

E. Discussion

Multiple transmitters: In a local wireless LAN, all devices should scramble the frames with same scrambling pattern. Otherwise, the eavesdropper may filter out unscrambled frames based on MAC address or radiometric fingerprints [48], [49]. Furthermore, if different devices use different scrambling patterns, *e.g.*, one device is emulating the walking and another is emulating running, the attacker may detect the difference and further recover the real activity pattern by discovering the common patterns among all transmitters. To address this problem, all devices in the same area should use the same scrambling parameters, and acknowledge frames should use the same scrambling vector as the data frame. This can be done by sharing the security key through a secure channel during the association process. Further study of this issue is out of the scope of this paper.

Attackers with multiple antennas: The attacker may use multiple antennas to monitor the channel state and use the

CSI-ratio approach by dividing the measurement on the same subcarrier of different antennas [16]. In this case, the attacker may cancel the scrambling of $S(f, t)$ similarly as discussed in Section IV-C to get:

$$\frac{H'_1(f_1, t)}{H'_2(f_1, t)} = \frac{H_1(f_1, t) \times S(f_1, t)}{H_2(f_1, t) \times S(f_1, t)} = \frac{H_1(f_1, t)}{H_2(f_1, t)} \quad (13)$$

where $H_1(f_1, t)$ and $H_2(f_1, t)$ are the channel response at antenna 1 and 2 on the same subcarrier of f_1 . In this case, there could be activity information leakage in the CSI-ratio. To address this issue, we can transmit the scrambled frames with randomized delays $\tau_i(t)$ at different transmission antennas, *e.g.*, scrambled by $S_1(f_1, t)$ at antenna 1 and scrambled by $S_2(f_1, t)$ at antenna 2. In this case, the received channel state estimation at antenna 1 of the attacker will be $H'_1(f_1, t) = H_{11}(f_1, t) \times S_1(f_1, t) + H_{21}(f_1, t) \times S_2(f_1, t)$, where $H_{11}(f_1, t)$ and $H_{21}(f_1, t)$ will be the CFR between transmitting antenna 1 and 2 to the receiving antenna 1. Therefore, eavesdroppers can no longer use the CSI-ratio approach to discover the real channel state when they have no knowledge on $S_1(f_1, t)$ and $S_2(f_1, t)$.

V. SYSTEM IMPLEMENTATION

We implemented CSS on the Xilinx ZCU-102 FPGA platform equipped with an AD-FMComms4-EBZ (FMC) as the RF frontend using the OpenWiFi framework. Our implementation is based on the 802.11a protocol and uses a bandwidth of 20 MHz. In addition, we modify the NIC driver and the PHY of the full network stack so that CSS can work as a normal wireless NIC and directly interact with commercial Wi-Fi devices. The overview of our system implementation is shown in Fig. 2.

NIC driver: Our NIC driver implementation generates the scrambling vectors using the procedures described in Section IV and stores them in two buffers, *i.e.*, buffer A and buffer B. While buffer A is being used by the scrambling vector selection function, buffer B is updated, and vice versa. The scrambling vectors are generated at a rate of 10KHz in the user space of the operating system, making it easy to incorporate other scramble algorithms. And the buffers' memory in the kernel space is shared between the scrambling vector selection function and a dedicated buffer management module. These designs ensure that the generation of scrambling vector and the update of the buffers does not introduce delays to the transmission process of the MAC layer in an underloaded multi-core system.

The driver selects a scrambling vector for each PLCP Service Data Unit (PSDU) based on the internal clock with a time granularity of 0.1 milliseconds. For 802.11a with 20 MHz bandwidth, we use a scrambling vector length of 104 Bytes, with one complex number encoded in two Bytes for each of the 52 subcarriers. As the scrambling vector is small, we send it along with the PSDU to the PHY layer via Direct Memory Access (DMA).

For obfuscating, implementing the selection of scrambling vector for each frame in the driver is sufficient. Furthermore, for secure sensing, where precise time synchronization via

TSF is needed, taking Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) into consideration makes implementing it in the PHY layer a better choice. However, given the complexity of the PHY implementation, we currently showcase the scrambling and descrambling CSI by implementing this function in the MAC layer. The MAC layer will encode the timestamp in the PSDU to help CSI descrambling, when there is an on-going secure sensing session. That is, we add a timestamp field to the standard format of the frame, therefore the receiver need to understand the non-standard frame format to perform the descrambling in the MAC layer. If the transmitter does not want to reveal the real CSI to the Wi-Fi sensor, it can just remove the timestamp and withhold the key $\mathcal{K}$, and the obfuscating still work.

PHY Layer: The PHY layer in CSS is implemented in the FPGA hardware. The PHY layer first extracts the per-frame scrambling vector from the PSDU. Then, the PHY multiplies each frequency domain OFDM symbol with the scrambling vector (SV) before the IFFT stage of the basic OFDM modulation process. We combine the multiplication operation with the existing serial-to-parallel operation to avoid extra delay.

For constant-valued preambles, such as STF and LTF symbols, the original modulation pipeline can store the time domain value of them in on-chip memory. However, to scramble these preambles, we need to perform the multiplication in the frequency domain so that we can only store the frequency domain value of the preambles in on-chip memory. This leads to an extra IFFT step for scrambling the preamble symbols. After that, CSS acts as a normal Wi-Fi transmitter.

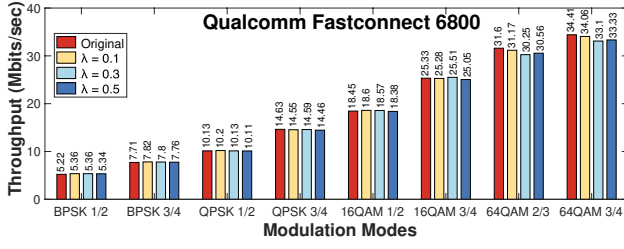
VI. EXPERIMENTAL RESULTS

A. Experiment Setup

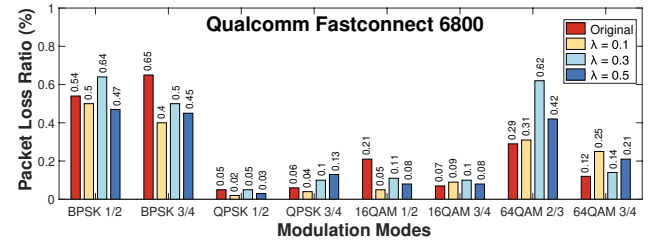
Our evaluations of communication performance are conducted in a busy open lobby with a size of 8m×12m. We use our CSS implementation on OpenWiFi framework as the transmitter and commercial WLAN NICs, including Intel AC3168 and AX200 on laptops, Qualcomm FastConnect 6200 and 6800 on smartphones, as receivers. The distance between the transmitter and receiver is 3 meters or 10 meters. Because WiFi-based sensing system does not work well in busy environment innately, we evaluate the scrambling and secure sensing performance elsewhere, that is a seminar room with a size of 4m×7m. The CSI is measured using the gr-ieee802-11 framework [50] on the USRP B210 software radio platform. We decode the PSDUs and extract the timestamp corresponding to each CSI sample if there is an on-going secure sensing session. Because of the low transmission power of the FMC, we use an RF amplifier with identical configurations in all experiments to enhance long-range communication and sensing performance of the OpenWiFi platform.

B. Communication Performance

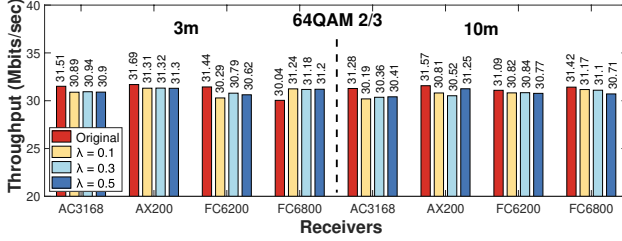
To validate our scrambling scheme does not negatively impact the communication performance, we conducted extensive experiments with commercial NICs under different scrambling



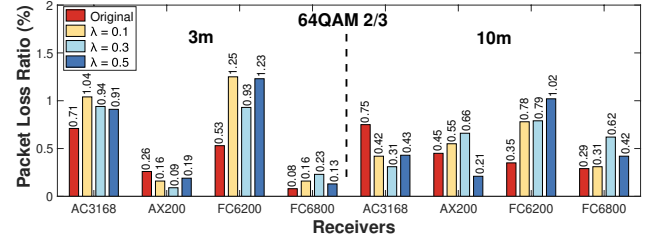
(a) Throughput with different modulation modes



(b) Packet loss ratio with different modulation modes

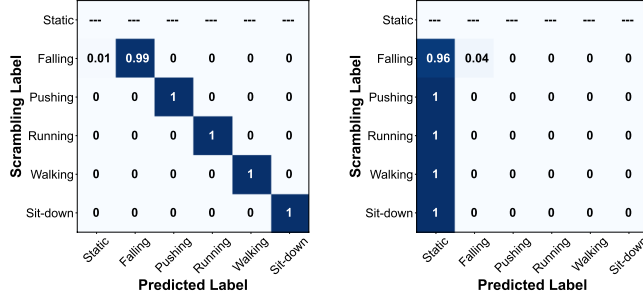


(c) Throughput with different receivers and distances

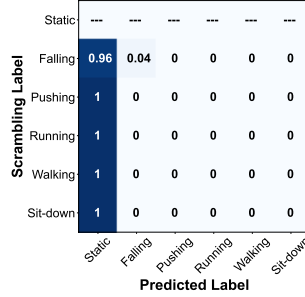


(d) Packet loss ratio with different receivers and distances

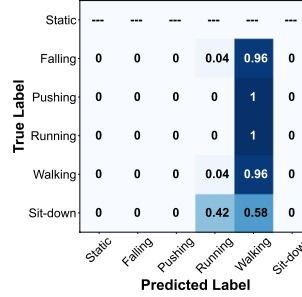
Fig. 6. Communication performance on different modulation modes, receivers and distances.



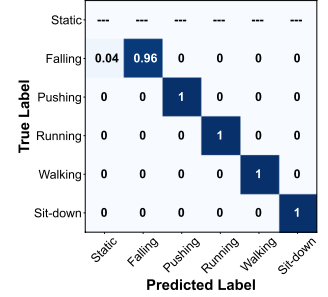
(a) Scrambling on static env.



(b) Descrambling on static env.



(c) Scrambling with walking



(d) Descrambling with walking

Fig. 7. Confusion matrices on scrambled/descrambled samples.

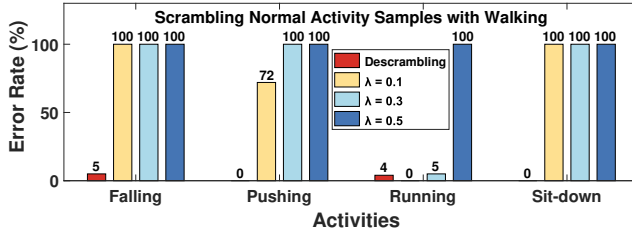


Fig. 8. Performance of scrambling physical activities with walking using different scrambling intensities.

intensities and different modulation modes at different distances. The end-to-end throughput and packet loss ratio (PLR) are used as key evaluation metrics. We measure the end-to-end throughput and PLR using iPerf, as our implementation on OpenWiFi can act as a normal access point and communicate with commercial WLAN NICs. In Fig. 6, we compare the communication performance in UDP of CSS with the original OpenWiFi. We observe that the throughput and PLR remain at the same level when scrambling is turned on at different levels of intensity that is indicated by the parameter λ . The communication performance scarcely degrades under different modulation modes, different distances and different types of commercial WLAN NICs. We also conduct a brief summary experiment to evaluate TCP performance using iPerf under

adaptive modulation mode at a scrambling intensity of 0.5 and the receiver is moving around. The result is 16.7 Mbits/sec, which is same as the original OpenWiFi.

C. Performance of Obfuscating

In this section, we evaluate the performance of CSS when an attacker is monitoring the channel and trying to recover the surrounding human activities. The attacker uses a software radio to capture the CSI and uses an algorithm similar to CARM [3] to recognize human activities. The activity set includes six categories: static environments, walking, running, falling, pushing, and sitting down. The attacker is trained using the CSI collected from real human activities of three users, with 120 samples per activity and one quarter of them are used as the testing set. The attacker achieves 99.5% and 98.6% accuracy on the training and testing sets respectively.

In the first scenario, the surrounding environment is static and CSS emulates different activities with LPC coefficients extracted from real human activities, 100 samples per activity in each scrambling intensity. Fig. 7(a) shows the confusion matrix of recognition results of the attacker, under a scrambling intensity of $\lambda = 0.5$. We observe that CSS successfully misleads the attacker to perceive the corresponding activity when there is no activity around. The overall success rate is

99.8% and 96.5% when the scrambling intensity is set to 0.5 and 0.3, respectively. With a smaller scrambling intensity, *e.g.*, $\lambda = 0.1$, CSS has a low success rate in misleading the attacker. This is mainly due to the power of the artificial activity signal is not large enough, such that it is submerged by the noise in CSI.

In the second scenario, we ask users to perform different activities in the set individually and the transmitted frames are scrambled by the walking activity with different intensities. Fig. 7(c) shows the confusion matrix of recognition results of the attacker when the physical activity is scrambled by emulated walking with a scrambling intensity of $\lambda = 0.5$. We observe that in most cases the attacker infers that there is someone walking in the environment, regardless of what user is actually doing. This is because the variations of channel state due to real human body's movement are obfuscated by the artificial channel state variations, as shown in Fig. 5(c). However, when the scrambling intensity is reduced to 0.3 or 0.1, the attacker can recognize the running activity correctly, as shown in Fig. 8. This is because the power of emulated walking is small compared to the physical running signal so that the spectrogram still contains a major component of running.

In the third scenario, we evaluate the performance of obfuscating breath signals. We ask the user to sit and breathe normally at a distance of 2 meters to the transmitter and receiver. The channel is scrambled by the pushing activity with a scrambling intensity of 0.3. We use an algorithm similar to [51] to estimate the breath rate of the user. The estimation error of the attacker increase from an average of 0.25 BPM (Breaths Per Minute) to 11.6 BPM, because the movement of pushing is similar to the expansion and contraction of the chest on a larger scale.

D. Secure Sensing

We evaluate the secure sensing capability of legitimate sensors by using the secret key $\mathcal{K}$ with the timestamp corresponding to the frame to generate the same set of scrambling vectors to recover the real channel state from the scrambled CSI data. The captured CSI samples in the three scenarios described in Section VI-C are also used in this section, and the sensing models of legitimate sensors are same as the attackers in Section VI-C. For the first scenario, Fig. 7(b) shows the recognition result of the recovered channel state for all scrambling intensities. The legitimate sensor successfully perceives the environment is static with an accuracy of 99.3%. For the second scenario, Fig. 7(d) shows the recognition result of the recovered channel state. The legitimate sensor successfully perceives the true activities in an accuracy of 99.3%. For the third scenario, the average error of legitimate sensor is 0.33 BPM, which is at the same level as the original setup, *i.e.*, 0.25 BPM. These results indicate that legitimate sensors are able to recover the physical activity from scrambled signals, even for weak signals such as breath.

E. System Overhead

CPU Time: The time complexity for generating pseudo-random numbers using the secret key $\mathcal{K}$ is $O(n)$. As shown by Eq. (8) and (12), the LPC processing also has a linear time complexity. The Hilbert transform, which is utilized for emulating the phase variations at the center subcarrier, has a time complexity of $O(n \log n)$. The Hilbert transform is not implemented at a generation rate of 10,000 samples per second, but at 800 samples per second. Subsequently, a linear resampling algorithm with a time complexity of $O(n)$ is applied. The scrambling vector buffer is updated every 10 seconds. Our LPC-based signal generator is implemented in Python and runs in the user space of Linux on the ZCU-102 platform, which is equipped with a quad-core ARM Cortex-A53 CPU. It takes 1.16 seconds to generate 100,000 scrambling vectors and takes 0.6 *ms* to store them into the buffer.

The packet transmission process in the driver has extra computational cost due to the scrambling vector selection. The driver is unconcerned about the content in the buffer and simply fetches a scrambling vector from designated buffer position based on the system time within a very brief interval, specifically 70 *ns*. Furthermore, the time overhead of DMA transfer is negligible for the driver, given its asynchronous nature.

FPGA Resource: In accordance with the utilization report generated by Vivado Design Suite 2018.3, after integrating our CSS into the OpenWiFi framework on the ZCU-102 FPGA platform, the additional consumption of FPGA resource is shown in TABLE I. The consumed resources, as listed, do not pose a bottleneck even on other resource-constrained platforms.

TABLE I
FPGA RESOURCE CONSUMPTION

	LUT	LUTRAM	FF	BRAM	DSP
Quantity	2420	400	2155	0.5	8
Percentage	0.88	0.28	0.39	0.05	0.32

Impact on PAPR: The larger Peak-to-Average Power Ratio (PAPR) of the signal, the higher performance requirements of the power amplifiers and RF chains for the transmitter. CSS scrambles the channel by multiplying the transmitted signal at different subcarriers with different complex values. To study the impact of CSS on PAPR, we perform simulations with randomly generated scrambling vectors using different scrambling intensities. As shown in Fig.9, we find that the scrambled symbols have identical PAPR distribution as original symbols in each order of modulation mode.

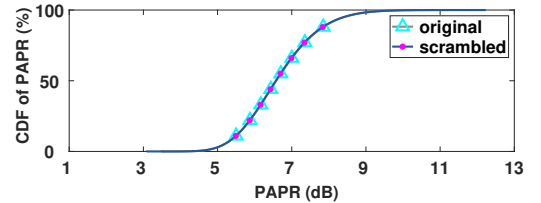


Fig. 9. Impact on PAPR distribution

VII. CONCLUSION

In this paper, we propose CSS, a general framework for obfuscating the Wi-Fi channel with LPC-based random scrambling vectors. Our implementation on a real communication system and extensive experiments in real-world scenarios show that CSS can provide always-on protection against channel state leakage without disturbing normal communications, and preserve the sensing ability of legitimate sensors. We believe that our approach can be extended to a wide range of wireless systems that use the OFDM modulation scheme.

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